

Writing to Files in C

In C, writing data to a file is done using file handling functions from the `<stdio.h>` library. This guide covers the essential methods to store data persistently on a disk.

1. File Pointer Declaration

A file is accessed using a specialized pointer type called **FILE**.

```
FILE *fp;
```

Pro Tip: The pointer stores the address of a structure that contains information about the file (buffer, current position, etc.).

2. Open File in Write Mode

To write to a file, open it using `fopen()`. Choosing the correct mode is critical:

- `"w"` (Write): Erases existing content and starts fresh. If the file doesn't exist, it creates it.
- `"a"` (Append): Keeps existing data and adds new data at the very end.

```
fp = fopen("data.txt", "w");
```

Always verify the pointer to prevent system crashes.

3. Writing Data to File

a) Using fprintf()

This is the most flexible method as it allows formatted output (like `printf`).

```
fprintf(fp, "Score: %d\n", 100);
```

b) Using fputs()

Used specifically for writing strings without formatting.

```
fputs("Welcome to CodeHive Learning.", fp);
```

c) Using fputc()

Writes a single character to the file. This is useful for character-by-character processing.

```
fputc('A', fp);
```

4. Closing the File

The **fclose(fp)** function is mandatory. It clears the buffer and ensures data is actually written to the disk.

Simple Example

```
#include <stdio.h>

int main() {
    FILE *fp;
    fp = fopen("file.txt", "w");

    if (fp == NULL) {
        printf("Error: File not opened");
        return 1;
    }

    fprintf(fp, "Writing data to file in C\n");
    fputs("CodeHive Documentation", fp);

    fclose(fp);
    return 0;
}
```

Comparison Table

Mode	Description	Existing Data
"w"	Write	Overwritten
"a"	Append	Preserved

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Problem

Developer Best Practices

- **Check for NULL:** Always validate `fp` before writing.
- **Buffering:** Large writes are stored in a buffer; use `fflush()` if manual sync is needed.
- **Paths:** Use double backslashes for Windows paths (e.g., "C:\\temp\\data.txt").



Key Points (Very Important ★)

- Use **<stdio.h>** for file handling.
- **"w"** overwrites file data entirely.
- **"a"** adds data without deleting old content.
- Always check **fp != NULL**.
- **Always close the file** using **fclose()**.

Thank you for learning with CodeHive!